1. Frobenius-orbit index calculus on Koblitz curves, fully priced, against the best rho

2026-10-14 · Conductor T-1 · Class: **engineering** · Study `research/ecbench_frobenius_orbit_vs_rho_20261014` · Protocol frozen in commit `e1708eef8` before claim data.

Verdict: constant-only. With every native priced, Frobenius-orbit IC ($m = 2$, `compact-orbit-scan` + `mitm-frobenius-counted`, `stride-renew` targets) is not below rho-frob on any of eight Koblitz curves ($\log_2 r$ = 11–39). The paired ratio grows with r: slope $\beta = 0.165[0.117, 0.213]$ per $\ln r$ with $\ln n$ in the fit (model: $\frac{1}{6}$). Orbits are 2–15× cheaper than the subspace base (constant), and beat rho-strong only where rho-strong's fixed set-up dominates ($\log_2 r \leq 17$). No security claim.

1.1. What was built

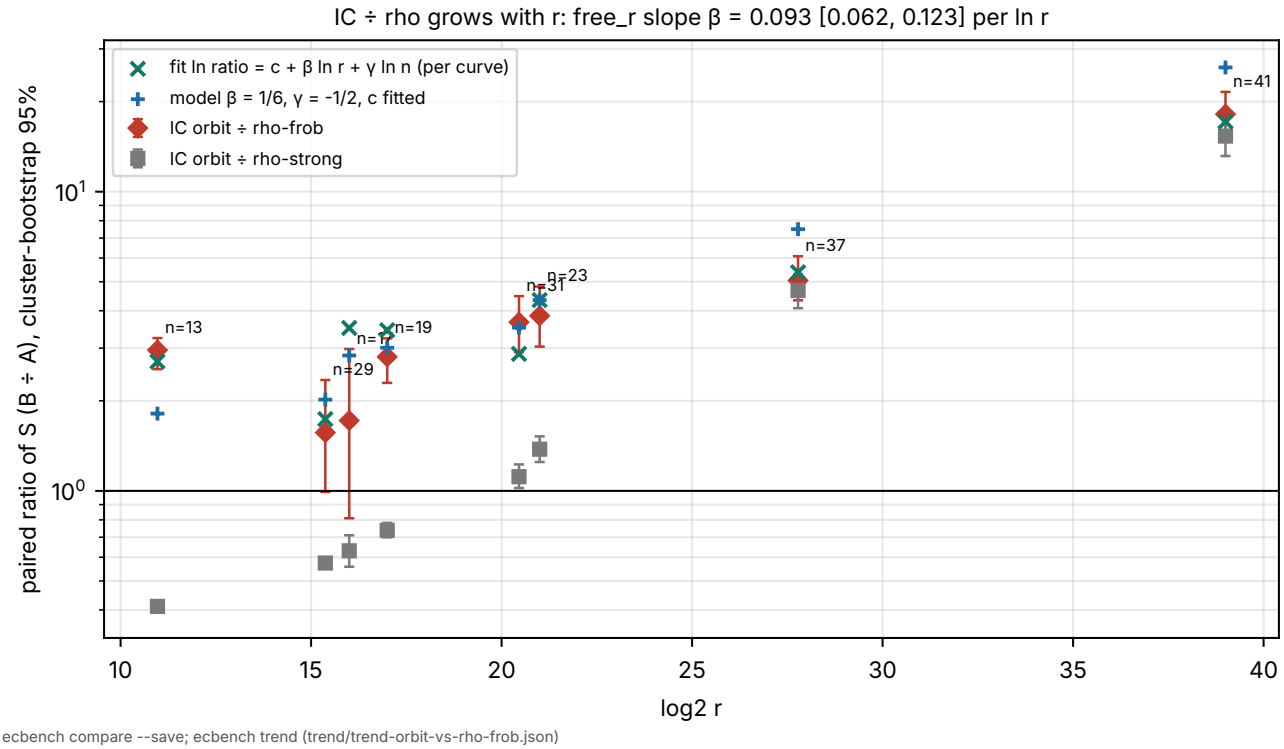
- **Full pricing.** New opt-in methods `ic.pipeline_full`, `rho.signed_frobenius_full`, `rho.signed_frobenius_strong_full` run unchanged solvers, then price every native counter from `src/cryptanalysis/ecbench/full_prices.json` (ratios to one affine addition measured by `ecbench price-units`, median of 15 passes, this host). Unknown counters stay unpriced. Existing methods and `docs/ic/calibration.json` unchanged; committed sessions replay as before.
- **`ecbench trend`**: weighted least squares of ln(paired ratio) on $\ln r$ and $\ln n$ from `compare --save` files.

1.2. Panel and verification

Eight sessions, 16 fresh public targets × 3 rounds (+1 warm-up), L0, 4 vCPU Xeon 2.80 GHz shared. 2,816/2,816 records verified; 2,112/2,112 measured records replayed identically; A/A exactly 1.000 on every curve; no record a lower bound.

curve	$\log_2 r$	h	c	orbit ÷ rho-frob	orbit ÷ rho-strong	renew ÷ walk	orbit ÷ negfold
a0 n13	10.97	4	1	2.956 [2.552, 3.242]	0.411 [0.397, 0.423]	0.403 [0.395, 0.412]	0.511 [0.465, 0.562]
a1 n17	16.00	2	1	1.719 [0.810, 2.980]	0.631 [0.558, 0.711]	0.503 [0.448, 0.562]	0.234 [0.203, 0.270]
a0 n19	17.00	4	2	2.805 [2.295, 3.236]	0.739 [0.699, 0.782]	0.556 [0.525, 0.591]	0.148 [0.131, 0.170]
a0 n23	21.00	4	3	3.847 [3.035, 4.819]	1.379 [1.249, 1.521]	0.753 [0.678, 0.834]	0.065 [0.059, 0.071]
a1 n29	15.37	12646	1	1.566 [0.992, 2.347]	0.574 [0.556, 0.593]	0.480 [0.464, 0.499]	–
a0 n31	20.46	1492	2	3.664 [2.907, 4.476]	1.116 [1.022, 1.226]	0.668 [0.607, 0.733]	–
a0 n37	27.78	596	8	5.048 [4.335, 6.086]	4.676 [4.080, 5.310]	1.029 [0.903, 1.176]	–
a0 n41	39.00	4	90	18.16 [15.67, 21.55]	15.34 [13.16, 18.28]	1.009 [0.938, 1.082]	–

Paired B ÷ A ratios of S from `ecbench compare`, cluster-bootstrap 95 % intervals. h = cofactor, c = orbit columns.



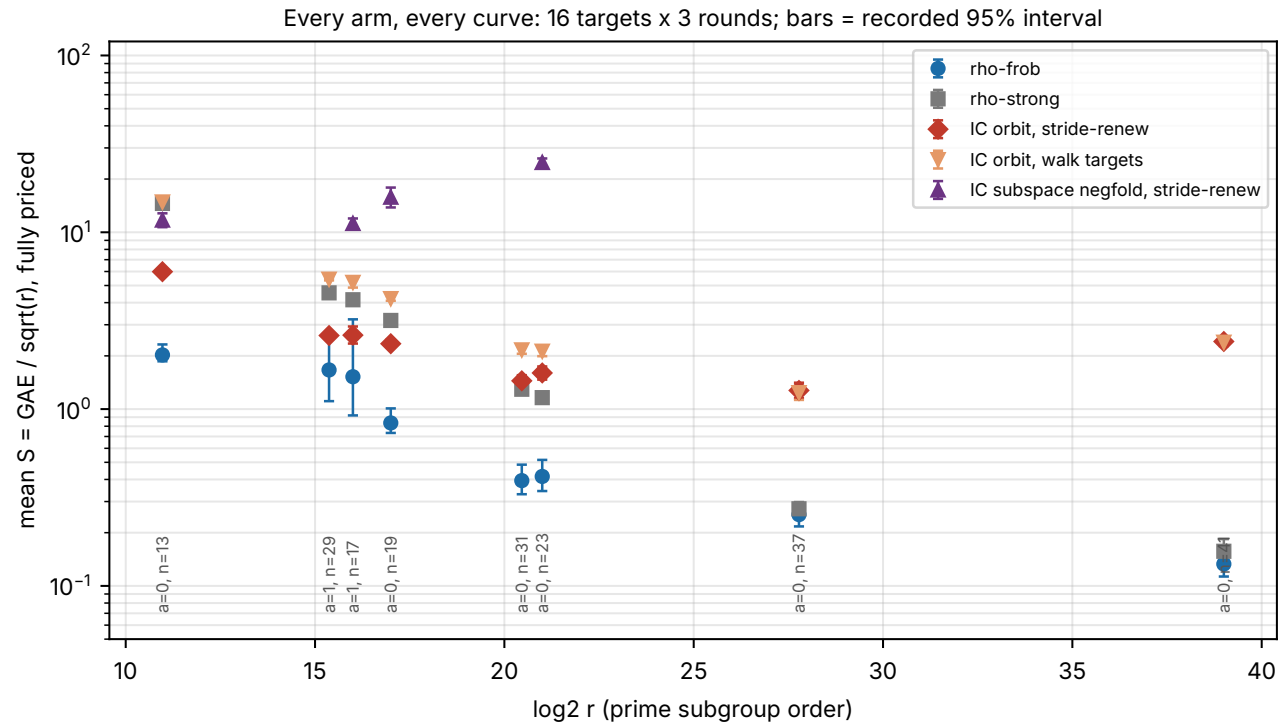
ecbench compare --save; ecbench trend (trend/trend-orbit-vs-rho-frob.json)

Figure 1: Paired ratios against $\log_2 r$ with the `free_r_n` and fixed-model fitted values. Sessions as listed in README; `trend/trend-orbit-vs-rho-frob.json`.

1.3. Trend (H6)

fit	coefficients (95 %)	χ^2/dof
$c + \beta \ln r$	$\beta = 0.093[0.062, 0.123]$	9.9
$c + \beta \ln r + \gamma \ln n$	$\beta = 0.165[0.117, 0.213]$, $\gamma = -1.18[-1.90, -0.46]$	3.9
model $c + \ln r/6 - \ln n/2$	$e^c = 1.84[1.41, 2.40]$	15.7

$\beta > 0$ in both free fits: constant-only by the frozen rule. The fixed model fails the χ^2/dof ≤ 2 rule; the free γ rests on the two large-cofactor small-r curves where c is at its integer floor and rho-frob is heavy-tailed, so it is descriptive. Extrapolating `free_r_n` to $r \approx 2^n$ (derived, not measured): the ratio is smallest near $n \approx 10$ at about 3.3 and never below 1.

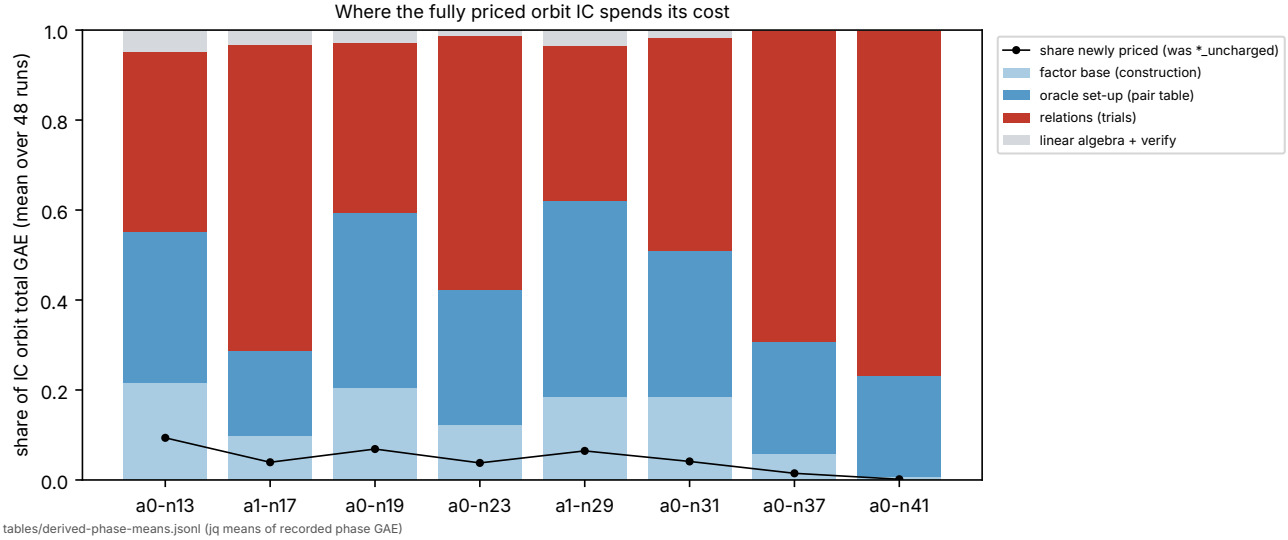


Sessions: research/ecbench_frobenius_orbit_vs_rho_20261014/sessions/*; ecbench table

Figure 2: Mean S of every arm on every curve with recorded 95 % intervals (`ecbench table`).

1.4. Pricing

The orbit base's formerly `*_uncharged` construction work costs 9.4 %, 6.9 %, 3.8 % and 0.16 % of the IC total at n = 13, 19, 23, 41: the old lower bound hid little. All natives are 36 % (n = 13) to 16 % (n = 41) of the IC total; hash probes are priced on a 2^{20}-entry map, about 11× the pinned ratio, which leans against IC. With every native priced at zero on both sides (derived, ratio of means), IC ÷ rho-frob is 1.98, 1.28, 1.84, 2.77, 0.96, 2.60, 4.09, 16.3 in panel order, so the verdict does not hinge on prices.



tables/derived-phase-means.jsonl (jq means of recorded phase GAE)

Figure 3: Phase shares of the orbit-renew arm (means of recorded phase GAE, `tables/derived-phase-means.jsonl`).

1.5. Limits

Toy and medium sizes on one shared host (L0; wall descriptive); prices are one host's microbenchmark ratios; c and d chosen on exploratory seeds (disclosed); negfold only at n ≤ 23 with cofactor ≤ 4; rho exponents fitted on these sizes are pre-asymptotic (rho-frob α = 0.238 [0.135, 0.303]); two a1 sessions record `git_dirty` from a whitespace-only `rustfmt` during the run (binary built from `e1708eef8`, unchanged).